A Response-Blind Holdout

for the Finite $c = 1$ Bandwidth Rule

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Abstract

TPC-375 found that the first member of a finite block-band menu matching its high-Q spectral failure support was the block-distance cutoff $c = 1$. This paper tests that rule on a response-blind grid-index holdout. We use the three reserved indices $(5, 15, 30)$ of the previously declared grid $a_j = 1010001 + 401j$, with count 2048, eight blocks of length 256, $Q \in \{512, 2048, 8192\}$, beta 2, exponent one, and the all-plus law. On the complete nine-row panel, the $c = 1$ band has spectral failure profile $(0, 3, 3)$ by increasing Q, exactly the parent profile, and has no Schur-cap failure. The selected full-mode band Rayleigh retention is between 0.93760019185559207 and 0.976941204869197. The result is a finite grid-index replication only: the two lower-index holdout windows overlap nearby training windows, and no origin-uniform, growing, arithmetic, or twin-prime conclusion is claimed.

1 Question and claim boundary

The recent finite TPC audits locate a recurring high-Q spectral signal in a normalized prime-shell matrix. TPC-374 showed that a predeclared near-block band with block distance at most three reproduced six parent failure keys. TPC-375 then compared the nested cutoffs $c = 0, 1, 2, 3$ on the same complete beta-2 panel. Its first parent-support match was $c = 1$. The natural next question is transfer to origins that were reserved by the earlier origin-grid protocol rather than used in that panel.

The present result is deliberately finite. The term *holdout* means a set of grid indices fixed independently of the signed response; it does not mean that all coordinate intervals are disjoint. In particular, the first two holdout intervals overlap a neighboring training interval by a small number of coordinates. We therefore do not interpret this experiment as an independent physical source sample.

2 Finite normalized object

For a frozen interval $I = [a, a + 2047] \cap \mathbb{Z}$, $Q \in \{512, 2048, 8192\}$, and a prime $p \in (Q, 2Q]$, put

$$K_p(u,t) = p\left(\frac{p}{Q}\right)^2 \frac{66^2}{66^2 + (u-t)^2}\left(\mathbf{1}_{p|(u-t)} - \frac{1}{p-1}\right)\mathbf{1}_{u\neq t}\mathbf{1}_{p\nmid u}\mathbf{1}_{p\nmid t}. \tag{1}$$

The all-plus numerator and its row geometry are

$$A(u,t) = \sum_{Q<p\leq 2Q} K_p(u,t), \qquad G(u) = \sum_{t\in I}\sum_{Q<p\leq 2Q} K_p(u,t)^2. \tag{2}$$

The full normalized matrix is

$$\mathsf{T}(u,t) = \frac{A(u,t)}{\sqrt{G(u)G(t)}}. \tag{3}$$

Write $b(u) = \lfloor (u-a)/256 \rfloor$. The frozen $c = 1$ band and tail are

$$\mathsf{B}(u,t) = \mathsf{T}(u,t)\mathbf{1}_{|b(u)-b(t)|\leq 1}, \qquad \mathsf{R} = \mathsf{T} - \mathsf{B}. \tag{4}$$

The geometry is a finite sum of rational squares. At the exact anchor $I = [1012006, 1012019)$ and $Q = 4$, direct rational arithmetic verifies symmetry and strict positivity. Also, entrywise $\mathsf{T} = \mathsf{B} + \mathsf{R}$. Thus, if $\mathsf{T}v = \lambda v$ and $\|v\|_2 = 1$, then

$$v^{\mathsf{T}}\mathsf{B}\,v + v^{\mathsf{T}}\mathsf{R}\,v = v^{\mathsf{T}}\mathsf{T}v = \lambda. \tag{5}$$

These are finite identities and carry no assertion about a growing family.

3 Response-blind holdout protocol

The earlier candidate grid is

$$a_j = 1010001 + 401j, \qquad 0 \leq j < 41.$$

TPC-375 used the training indices $(0, 20, 40)$. Before reading any holdout metric, this audit reserved $(5, 15, 30)$, giving

$$(a_5, a_{15}, a_{30}) = (1012006, 1016016, 1022031).$$

The complete Cartesian product of these origins and the three Q anchors is evaluated. No origin, row, cutoff, eigenmode, or sign is selected after a response is observed. The full mode is the eigenvalue of largest absolute value, with the minimum eigenvalue winning an exact tie.

Table 1: Frozen protocol and finite census.

quantity	value
holdout indices	$(5, 15, 30)$
holdout origins	$(1012006, 1016016, 1022031)$
window and blocks	2048 points; 8×256
shell anchors	$Q = 512, 2048, 8192$
beta, exponent, law	$2, 1,$ all-plus
band	block distance ≤ 1
spectral failures by Q	0/3, 3/3, 3/3
Schur failures	0/9

4 Results

The $c = 1$ band spectral values range from 0.50281931 to 0.50283444 at $Q = 512$, from 0.66562826 to 0.66563868 at $Q = 2048$, and from 0.66694246 to 0.66694503 at $Q = 8192$; the certificate records the full precision. Relative to the cap 0.64, this gives six failures, with the same $(0, 3, 3)$ Q-profile as the TPC-375 panel. The Schur cap 0.83 is not crossed in any row.

Table 2: Row-level band values and selected-mode tail summary.

origin	Q	band spectrum	band Schur	abs. retention
1012006	512	0.50283444162621627	0.53128395220859004	0.976941204869197
1012006	2048	0.66562917307210956	0.67846748108299049	0.93760084447589365
1012006	8192	0.66694245634594918	0.67972169418368655	0.93760019185559207
1016016	512	0.50282578731047578	0.53133693522854253	0.97693928706874245
1016016	2048	0.66563867656808429	0.67852090333711979	0.93760180369522450
1016016	8192	0.66694481889731727	0.67971862044077846	0.93760051104122999
1022031	512	0.50281930544856424	0.53129871974955478	0.97693949932444424
1022031	2048	0.66562825618491772	0.67848910457776990	0.93760054654011304
1022031	8192	0.66694502781223552	0.67973329833486718	0.93760019630794067

The selected full-mode band absolute-Rayleigh retention is

$$0.93760019185559207 \leq \frac{|v^{\mathsf{T}} \mathsf{B}\, v|}{|\lambda|} \leq 0.976941204869197,$$

and the maximum corresponding absolute tail fraction is 0.062399808144408715. The producer checks eigenvector residuals, norm, symmetry, and the finite Schur/Frobenius envelopes. A separate checker reconstructs the prime shell in descending order and obtains the same failure profile and principal metrics within its declared numerical tolerance.

5 Audit and reproducibility

The repository package contains a producer, a reverse-shell independent checker, a mutation stress test, an exact-anchor record, and a local fail-closed Bridge-B. The producer locks the TPC-375 engine and certificate by LF-normalized SHA-256; the independent checker does not import the producer. Normal and optimized Python runs are required to have empty standard error and byte-identical output.

The exact finite statements are the fixed grid protocol, the finite nonnegative-square geometry, the common normalization, the band/tail identity, and the Rayleigh decomposition. The numerical statements are scoped to the nine declared rows. The official Route-A and Route-B evaluator files named by the Session are absent, so no official evaluator pass is asserted.

6 Conclusion and next question

The c=1 bandwidth rule survives this predeclared grid-index holdout at the level of its Q-profile: six high-Q spectral failures and no Schur failures. This is a concrete finite transfer edge from TPC-375, while the overlap between some coordinate intervals is an explicit obstruction to stronger sample-independence language. Origin/window uniformity, cross-block causality, a growing operator estimate, source-uniform arithmetic L^2, fixed-power credit, and the twin-prime endpoint remain open.

The next minimal experiment is a predeclared window-scale/count holdout for the same c=1 rule, recorded in the route ledger as `TEST_C1_WINDOW_SCALE_HOLDOUT`.